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EDUCATION·

2015/09—2019/01 Beihang University School of Astronautics Doctor
Research interests: hyperspectral remote sensing processing, deep learning, machine learning.
Tutor: Professor Zhenwei Shi
Major: Pattern Recognition and Intelligent Systems

2013/09—2015/06 Beihang University School of Astronautics Master
Research interests: image preprocessing, remote sensing image target detection and recognition, active learning and semi-supervised learning.
Tutor: Professor Zhiguo Jiang
Major: Pattern Recognition and Intelligent Systems

2009/09--2013/06 Beihang University School of Astronautics Bachelor
Major: Detection guidance and control technology

·EXPERIENCE·

2024-present Full Professor, School of Statistics and Data Science, Nankai University

2019- 2024 Associate Professor, School of Statistics and Data Science, Nankai University
Research interests: remote sensing image processing, machine learning.
Undertaking courses: Data Mining, Machine Learning, Deep Learning

·PAPER·

I have published more than **70 research papers**, including **30 IEEE Transactions**, **4 ESI hot papers** and **7 ESI highly cited papers.** Published papers have been cited by Google Scholar for more than 2800 times. The list of some representative papers is as follows. (* is the corresponding author):

Year 2027

[1] Ziye Zhang, **Bin Pan*** and Zhenwei Shi. Radiative-Structured Neural Operator for Continuous Spectral Super-Resolution. IEEE Transactions on Multimedia, 2027

[2] Shiyu Shen, **Bin Pan***, Zongbin Wang and Zhenwei Shi. A Distribution-Free Efficient Bayesian Framework for Domain Generalization. IEEE Transactions on Multimedia, 2027

Year 2026

[3] **Bin Pan**, Shiyu Shen, Zongbin Wang, Zhenwei Shi and Xia Xu. Preserving Domain Generalization in Fine-Tuning via Joint Parameter Selection. IEEE Transactions on Multimedia, 2026. **DOI:** 10.1109/TMM.2026.3703329

[4] **Bin Pan**, Lan Ma, Ruiying Li, Xia Xu and Zhenwei Shi. Uncertainty Modeling for Domain Generalization in Cross-Scene Hyperspectral Image Classification. IEEE Transactions on Geoscience and Remote Sensing, 2026. **DOI:** 10.1109/TGRS.2026.3696517

[5] Ruiying Li, **Bin Pan***, Lan Ma, Xia Xu and Zhenwei Shi. Multitemporal Latent Dynamical Framework for Hyperspectral Images Unmixing. ISPRS Journal of Photogrammetry and Remote Sensing, 2026. **DOI:** 10.1016/j.isprsjprs.2026.04.048

[6] Junzheng Chu, **Bin Pan***, Tianyang Shi and Zhenwei Shi. Domain Generalization via Domain Uncertainty Shrinkage. Pattern Recognition, 2026. **DOI:** 10.1016/j.patcog.2026.113118

[7] Shiyu Shen, **Bin Pan***, Tianyang Shi, Tao Li and Zhenwei Shi. Be Bayesian by Attachments to Catch More Uncertainty. Pattern Recognition, 2026. **DOI:** 10.1016/j.patcog.2026.113084

[8] Ruiying Li, **Bin Pan***, Qiaoying Qu, Xia Xu and Zhenwei Shi. A Copula-Guided Temporal Dependency Method for Multitemporal Hyperspectral Images Unmixing. Pattern Recognition, 2026. **DOI:** 10.1016/j.patcog.2025.112971

[9] Zhe Gao, **Bin Pan*** and Zhenwei Shi. C3DG: Conditional Domain Generalization for Hyperspectral Imagery Classification with Convergence and Constrained-risk Theories. IEEE Transactions on Geoscience and Remote Sensing, 2026. **DOI:** 10.1109/TGRS.2026.3696512

[10] Qiaoying Qu, **Bin Pan***, Ruiying Li and Zhenwei Shi. Implicit Continuous Full-spectrum Response Network for Visible-to-Infrared Image Translation. IEEE Transactions on Geoscience and Remote Sensing, 2026. **DOI:** 10.1109/TGRS.2026.3696523

[11] Mengze Wang, **Bin Pan*** and Haiyang Xu. AFS-Net: An Anchor-Free Domain Adaptation Network for SAR Ship Detection. IEEE Geoscience and Remote Sensing Letters, 2026. **DOI:** 10.1109/LGRS.2025.3646070

[12] Zongjun Han, Lu Bai, **Bin Pan** and Peng Ren. From Mutual Guide to Confucius Tri-learning: A Theoretical Justification. Pattern Recognition, 2026. **DOI:** 10.1016/j.patcog.2026.113447

[13] Haiyang Xu, Xia Xu, Yinghao Tan, Ziye Zhang, Nanhe Chen, **Bin Pan** and Zhenwei Shi. Loc-Edit: Localized Color Editing in Neural Radiance Fields. Neurocomputing, 2026. **DOI:** 10.1016/j.neucom.2026.133184

Year 2025

[14] Jiao Liu, **Bin Pan*** and Zhenwei Shi. CR-Famba: A Frequency-Domain Assisted Mamba for Thin

Cloud Removal in Optical Remote Sensing Imagery. IEEE Transactions on Multimedia, 2025. **DOI:** 10.1109/TMM.2025.3542976

[15] Ye Zhu, Jiaxin Han, **Bin Pan*** and Zhenwei Shi. DiffPR-Net: Few-Shot Remote Sensing Scene Classification Based on Generative Diffusion and Prototype Rectified Model. IEEE Transactions on Geoscience and Remote Sensing, 2025. **DOI:** 10.1109/TGRS.2025.3528071

[16] Shiyu Shen, **Bin Pan***, Yixin Wang, Yihang Liu and Zhenwei Shi. A Second Order Stationarity based Confidence Assessment Method for Temperature Forecast, IEEE Geoscience and Remote Sensing Letters, 2025. **DOI:** 10.1109/LGRS.2025.3533575

[17] Jun Zhang, Zhang Xupeng, Shuai Liu, **Bin Pan***, Zhenwei Shi. FIE-Net: Foreground Instance Enhancement Network for Domain Adaptation Object Detection in Remote Sensing Imagery. IEEE Transactions on Geoscience and Remote Sensing, 2025. **DOI:** 10.1109/TGRS.2025.3557171

[18] Wei Liu, Haixia Bi, Xuehu Zhu, Chen Xu and **Bin Pan**. Unsupervised and Unpaired Fusion-Based Hyperspectral Image Super-Resolution Based on Reflectance Migration. IEEE Transactions on Geoscience and Remote Sensing, 2025. **DOI:** 10.1109/TGRS.2025.3578150

Year 2024

[19] Junzheng Chu, **Bin Pan***, Xia Xu, Tianyang Shi, Zhenwei Shi and Tao Li. Joint Variational Inference Network for Domain Generalization. Pattern Recognition, 2024. **DOI:**10.1016/j.patcog.2024.110587

[20] Yuyou Gao, **Bin Pan***, Xia Xu,Xinyu Song and Zhenwei Shi. A Reversible Generative Network for Hyperspectral Unmixing with Spectral Variability. IEEE Transactions on Geoscience and Remote Sensing, 2024. **DOI:** 10.1109/TGRS.2024.3403926

[21] Jiao Liu, **Bin Pan*** and Zhenwei Shi. Cascaded Memory Network for Optical Remote Sensing Imagery Cloud Removal. IEEE Transactions on Geoscience and Remote Sensing, 2024. **DOI:** 10.1109/TGRS.2024.3376609

[22] Jiajun Xie, **Bin Pan***, Xia Xu and Zhenwei Shi. MiSSNet: Memory-inspired Semantic Segmentation Augmentation Network for Class-Incremental Learning in Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing, 2024. **DOI:** 10.1109/TGRS.2024.3360701

[23] Jun Zhang, Cheng Zhang, Shuai Liu, Zhenwei Shi and **Bin Pan***. Three-dimensional Frequency Domain Transform Network for Cross-Scene Hyperspectral Image Classification. IEEE Transactions on Geoscience and Remote Sensing, 2024. **DOI:** 10.1109/TGRS.2024.3491159

[24] Zhenning Shi, Haoshuai Zheng, Chen Xu, Changsheng Dong, **Bin Pan**, Xueshuo Xie, Along He, Tao Li and Huazhu Fu. Resfusion: Denoising Diffusion Probabilistic Models for Image Restoration Based on Prior Residual Noise. NeurIPS, 2024. **DOI:** 10.48550/arXiv.2311.14900

[25] Yinghao Tan, Junfeng Wu, Yihang Liu, Shiyu Shen, Xia Xu and **Bin Pan**. A Lightweight Transformer Based Spatiotemporal Analysis Prediction Algorithm for High Dimensional Meteorological Data. Remote Sensing, 2024. **DOI:**10.3390/rs16234545

Year 2023

[26] **Bin Pan**, Zhehao Xu, Tianyang Shi, Tao Li and Zhenwei Shi. An Imbalanced Discriminant Alignment Approach for Domain Adaptive SAR Ship Detection. IEEE Transactions on Geoscience and Remote Sensing, 2023. **DOI:** 10.1109/TGRS.2023.3303507

[27] Zhe Gao, **Bin Pan***, Xia Xu,Tao Li and Zhenwei Shi. LiCa: Label-indicate-Conditional-alignment Domain Generalization for Pixel-wise Hyperspectral Imagery Classification. IEEE Transactions on Geoscience and Remote Sensing, 2023. **DOI:** 10.1109/TGRS.2023.3300688

[28] Ruiying Li, **Bin Pan***, Xia Xu,Tao Li and Zhenwei Shi. Towards Convergence: A Gradient-Based Multiobjective Method With Greedy Hash for Hyperspectral Unmixing. IEEE Transactions on Geoscience and Remote Sensing, 2023. **DOI:** 10.1109/TGRS.2023.3267080

[29] Lukui Shi, Ruiyun Zhao, **Bin Pan***, Zhengxia Zou and Zhenwei Shi. Unsupervised Multi-Modal Remote Sensing Image Registration via Domain Adaptation. IEEE Transactions on Geoscience and Remote Sensing, 2023. **DOI:** 10.1109/TGRS.2023.3333889

[30] Yuexin Duan, Xia Xu, **Bin Pan**, Tao Li and Zhenwei Shi. UnDAT: Double Aware Transformer for Hyperspectral Unmixing. IEEE Transactions on Geoscience and Remote Sensing, 2023. **DOI:** 10.1109/TGRS.2023.3310155

[31] Lukui Shi, Qingrui Zhang, **Bin Pan***, Jun Zhang and Yuanchao Su. GlobalLocal and Occlusion Awareness Network for Object Tracking in UAVs. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2023. **DOI:** 10.1109/JSTARS.2023.3308042

[32] Yuyou Gao, **Bin Pan***, Xinyu Song and Xia Xu. Extended-Aggregated Strategy For Hyperspectral Unmixing Based On Dilated Convolution. IEEE Geoscience and Remote Sensing Letters, 2023. **DOI:** 10.1109/LGRS.2023.3297577

[33] Qiaoying Qu, **Bin Pa**n, Xia Xu,Tao Li and Zhenwei Shi. Unmixing Guided Unsupervised Network for RGB Spectral Super-Resolution. IEEE Transactions on Image Processing, 2023. **DOI:** 10.1109/TIP.2023.3299197

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[34] **Bin Pan**, Qiaoying Qu, Xia Xu and Zhenwei Shi. Structure-Color Preserving Network for Hyperspectral Image Super Resolution. IEEE Transactions on Geoscience and Remote Sensing, 2022. **DOI:** 10.1109/TGRS.2021.3135028

[35] Jun Zhang, Simin Li, Yongfeng Dong, **Bin Pan*** and Zhenwei Shi. Hierarchical Similarity Alignment for Domain Adaptive Ship Detection in SAR Images. IEEE Transactions on Geoscience and Remote Sensing, 2022. **DOI:** 10.1109/TGRS.2022.3227626

[36] Lukui Shi, Linyi Kuang, Xia Xu, **Bin Pan*** and Zhenwei Shi. CANet: Centerness-Aware Network for Object Detection in Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing, 2022. **DOI:** 10.1109/TGRS.2021.3068970

[37] Xia Xu, Xinyu Song, Tao Li, Zhenwei Shi and **Bin Pan**. Deep Autoencoder for Hyperspectral Unmixing via Global-Local Smoothing. IEEE Transactions on Geoscience and Remote Sensing, 2022. **DOI:** 10.1109/TGRS.2022.3152782

[38] Yanyi Wei, Xia Xu, **Bin Pan***, Tao Li and Zhenwei Shi. A Multi-objective Group Sparse Hyperspectral Unmixing Method with High Correlation Library. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2022. **DOI:** 10.1109/JSTARS.2022.3200693

[39] Jun Zhang, Pengli Wu, Xia Xu, Ming Han and **Bin Pan***. PCS-LSTM: A Hybrid Deep Learning Model for Multi-stations Joint Temperature Prediction based on Periodicity and Closeness, Neurocomputing, 2022. **DOI:** 10.1016/j.neucom.2022.06.015

[40] Jun Zhang, Ruofei Huang, Yan Li and **Bin Pan***.Oriented Ship Detection Based on Intersecting Circle and Deformable RoI in Remote Sensing Images. Remote Sensing, 2022.

DOI:10.3390/rs14194749

[41] Zhenjie Tang, Qing Xu, Pengfei Wu, Zhenwei Shi and **Bin Pan***. Feedback Refined Local-Global Network for Super Resolution of Hyperspectral Imagery. Remote Sensing, 2022. **DOI:** 10.3390/rs14081944

[42] Jun Zhang, Yue Liu, Pengfei Wu, Zhenwei Shi and **Bin Pan***. Mining Cross Domain Structure Affinity for Refined Building Segmentation in Weakly Supervised Constraints. Remote Sensing, 2022. **DOI:** 10.3390/rs14051227

[43] Chen Sun, Jing Zhou, Yuchi Ma, Yijia Xu, **Bin Pan** and Zhou Zhang. A Review of Remote Sensing for Potato Traits Characterization in Precision Agriculture. Frontiers in Plant Science, 2022. **DOI:** 10.3389/fpls.2022.871859

Year 2021

[44] Jun Zhang, Jiao Liu, **Bin Pan***, Zongqing Chen, Xia Xu and Zhenwei Shi. An Open Set Domain Adaptation Algorithm via Exploring Transferability and Discriminability for Remote Sensing Image Scene Classification. IEEE Transactions on Geoscience and Remote Sensing, 2021. **DOI:** 10.1109/TGRS.2021.3110060

[45] Xia Xu, **Bin Pan***, Zongqing Chen, Zhenwei Shi and Tao Li. Simultaneously Multi-objective Sparse Unmixing and Library Pruning for Hyperspectral Imagery. IEEE Transactions on Geoscience and Remote Sensing, 2021. **DOI:** 10.1109/TGRS.2020.3016941

[46] Enhai Liu, Yu Zheng, **Bin Pan***, Xia Xu and Zhenwei Shi, DCL-Net: Augmenting the Capability of Classification and Localization for Remote Sensing Object Detection. IEEE Transactions on Geoscience and Remote Sensing, 2021. **DOI:** 10.1109/TGRS.2020.3048384

[47] Peiliang Zuo, Han Zhang, Chen Wang, Hua Jiang and **Bin Pan**, Directional Target Localization in NLOS Environments Using RSSTOA Combined Measurements. IEEE Wireless Communications Letters, 2021. **DOI:** 10.1109/LWC.2021.3109787

[48] Lukui Shi, Ziyuan Wang, **Bin Pan*** and Zhenwei Shi. An End-to-End Network for Remote Sensing Imagery Semantic Segmentation via Joint Pixel and Representation Level Domain Adaptation. IEEE Geoscience and Remote Sensing Letters, 2021. **DOI:** 10.1109/LGRS.2020.3010591

Year 2020

[49] Jun Zhang, Jiao Liu, **Bin Pan***, Zhenwei Shi. Domain Adaptation Based on Correlation Subspace Dynamic Distribution Alignment for Remote Sensing Image Scene Classification. IEEE Transactions on Geoscience and Remote Sensing, 2020. **DOI:** 10.1109/TGRS.2020.2985072

[50] **Bin Pan**, Xia Xu, Zhenwei Shi, Ning Zhang, Huanlin Luo and Xianchao Lan. DSSNet: A Simple Dilated Semantic Segmentation Network for Hyperspectral Imagery Classification. IEEE Geoscience and Remote Sensing Letters, 2020. **DOI:** 10.1109/LGRS.2019.2960528

[51] Jun Zhang, Min Zhang, **Bin Pan*** and Zhenwei Shi. Semisupervised Center Loss for Remote Sensing Image Scene Classification. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2020. **DOI:** 10.1109/JSTARS.2020.2978864

[52] Jun Zhang, Min Zhang, Liukui Shi, Wenjie Yan and **Bin Pan***. A Multi-Scale Approach for Remote Sensing Scene Classification Based on Feature Maps Selection and Region Representation. Remote Sensing, 2020. **DOI:** 10.3390/rs11212504

Year 2019

[53] **Bin Pan**, Zhenwei Shi and Xia Xu. Analysis for the Weakly Pareto Optimum in Multi-Objective based Hyperspectral Band Selection, IEEE Transactions on Geoscience and Remote Sensing, 2019. **DOI:** 10.1109/TGRS.2018.2886853

[54] **Bin Pan**, Zhenwei Shi and Xia Xu. Multi-objective Based Sparse Representation Classifier for Hyperspectral Imagery Using Limited Samples, IEEE Transactions on Geoscience and Remote Sensing, 2019. **DOI:** 10.1109/TGRS.2018.2853268

[55] **Bin Pan**, Zhenwei Shi, Xia Xu, Tianyang Shi, Ning Zhang and Xinzhong Zhu. CoinNet: Copy Initialization Network for Multispectral Imagery Semantic Segmentation, IEEE Geoscience and Remote Sensing Letters, 2019. **DOI:** 10.1109/LGRS.2018.2880756

Year 2018

[56] **Bin Pan**, Zhenwei Shi and Xia Xu. MugNet: Deep Learning for Hyperspectral Image Classification Using Limited Samples. ISPRS Journal of Photogrammetry and Remote Sensing, 2018. **DOI:** 10.1016/j.isprsjprs.2017.11.003

[57] Xia Xu, Zhenwei Shi and **Bin Pan**. ℓ0-based Sparse Hyperspectral Unmixing Using Spectral Information and A Multi-objectives Formulation. ISPRS Journal of Photogrammetry and Remote Sensing, 2018. **DOI:** 10.1016/j.isprsjprs.2018.04.008

Year 2017

[58] **Bin Pan**, Zhenwei Shi and Xia Xu. Hierarchical Guidance Filtering Based Ensemble Classification for Hyperspectral Image. IEEE Transactions on Geoscience and Remote Sensing, 2017. **DOI:** 10.1109/TGRS.2017.2689805

[59] **Bin Pan**, Zhenwei Shi and Xia Xu. RVCANet: A New Deep Learning Based Hyperspectral Image

Classification Method. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2017. **DOI:** 10.1109/JSTARS.2017.2655516

[60] **Bin Pan**, Zhenwei Shi, Xia Xu and Yi Yang. Hashing based Hierarchical Feature Representation for Hyperspectral Imagery Classification. Remote Sensing, 2017. DOI: 10.3390/rs9111094

[61] **Bin Pan**, Zhenwei Shi, Zhenyu An, Zhiguo Jiang and Yi Ma. A Novel Spectral-Unmixing-Based Green Algae Area Estimation Method for GOCI Data. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2017. **DOI:** 10.1109/JSTARS.2016.2585161

Year 2016

[62] **Bin Pan**, Zhenwei Shi, Ning Zhang, Shaobiao Xie. Hyperspectral Image Classification Based on Nonlinear Spectral-Spatial Network. IEEE Geoscience and Remote Sensing Letters, 2016. **DOI:** 10.1109/LGRS.2016.2608963

·SOCIAL WORK·

- Associated Editor of the SCI-indexed journal <Pattern Recognition>
- Associated Editor of the SCI-indexed journal <Infrared Physics & Technology>
- Associated Editor of the SCI-indexed journal <Robotics and Autonomous Systems>

- Guest Editor of the SCI-indexed journals <Remote Sensing> and <Electronics>and<Journal of Imaging>
- Awarded the First Prize of Science and Technology Award of CICC
- Awarded the Second Prize for Teaching Achievements in the National Applied Statistics Professional Degree Graduate Education
- Selected as one of the "Hundred Young Academic Leaders" at Nankai University
- Council Member of the Information Innovation Technology Committee, China Institute of Communications
- Council Member of the Remote Sensing Committee, China Society of Image and Graphics
- Council Member of the Tianjin Association for Intelligent Science and Technology
- PC Member, Session Chair for over a dozen conferences